Milestone 3.3 — Product Consistency & Quality Testing

Team Code Blooded | GigPoint Hackathon, 22 August 2026 | Problem 2: AI Ad Generation | Client: Kaladhar

The client's hard requirement is that a generated advertisement must not misrepresent the product. This document sets out how that is tested, and reports what the tests actually found — including a video that failed and a threshold problem the testing exposed.

All numbers are produced by the pipeline in this repository. Nothing here is estimated, and where something has not been tested it is listed in section 11 rather than left implied.

1. What was actually tested

Product	Run	Source image	Videos	Own calibration	Gates run
Saree	run_m2_real	saree_source.jpg, 500x724	4	Yes (threshold 51.0)	Gate 1 + Gate 2
Kurti	run_20260822_183825	upload.png, 2500x2500	1	Added for this report (63.2)	Gate 1 only (single clip)

Two product types have been tested on real generated video: a saree and a kurti. No other garment type has been run end to end, and none is claimed. A second kurti run (run_20260822_182724) holds an identical repeat pass of the same clip and is not counted as an independent test.

2. What the system checks, and why

Every check below is implemented and runs on every video. All four fidelity metrics are computed **inside the garment mask only** — the background is supposed to change completely, so measuring it would drown the signal — and on **every 5th frame**, because the drift observed is transient.

Check	What it detects	Why it matters
Garment segmentation	Which pixels are the product	Everything else is measured inside this region
Colour ΔE2000	Dye shift in the garment palette	The failure a customer notices on arrival
CLIP identity	A different garment substituted	Catches wholesale replacement, not just tint
Weave / texture	Handloom rendered as flat fabric	Weave is what a handloom brand sells
Border / structure	Motif distorted or redrawn (SSIM)	Border and pallu are the identifying detail
Per-frame drift	Brief failures mid-clip	A single-frame check misses transient drift
Motion / flicker	Frozen clips and hard cuts	Both are unusable as an advertisement
Garment occupancy	Product shrinking in frame	A product ad needs a visible product
Gate 1 (Fidelity Lock)	Overall product drift	Blocks misrepresentation before delivery
Gate 2 (Diversity)	Near-duplicate outputs	Answers the repetitiveness complaint
Safe Cut	Recoverable clean sections	Salvages footage without laundering a bad clip

3. Test methodology

Per-frame, not per-clip. 31 of 153 frames are scored per video, at stride 5. The four component scores are combined into a Fidelity Score of 0–100 using fixed weights (colour 0.35, identity 0.30, weave 0.25, structure 0.10).

Thresholds are calibrated, not guessed. The scale is anchored by deliberately damaging the run's own source image in known ways — hue shifts (dye change), blur (weave collapse), desaturation — and scoring those through the identical pipeline. The pass mark is then built from a constructed limit clip. This is why the two products end up with different thresholds.

3.1 Metric validation against known degradation

Before trusting the weave metric on real video it was tested against damage of known severity. It must rise monotonically as texture is destroyed, and stay near zero when the same fabric is merely rescaled.

Variant applied to the source	Weave distance	High-freq. ratio	Expected
identical	0.0000	0.0088	zero
gaussian blur k=3	0.2423	0.0028	rising
gaussian blur k=5	0.3033	0.0013	rising
gaussian blur k=9	0.3506	0.0002	rising

gaussian blur k=15	0.3604	0.0001	rising
rescaled x1.5 (same fabric)	0.0255	0.0095	near zero
flat colour, no weave at all	0.3668	0.0000	maximum

Monotonic under increasing blur, saturating on flat colour, and near zero under rescaling. An earlier cosine-based version of this metric returned ~0.0001 for every frame and was replaced; a regression test now locks this behaviour in.

3.2 Algorithm-level test suite

20 automated tests cover the logic behind the checks: **20 passed, 0 failed.** They include validation accept/reject cases, the brief sampler's minimum-difference rule, weave monotonicity and rescale robustness, subscore bounds, the structure check refusing low-confidence matches, diversity selection across clusters, and five Safe Cut refusal cases. These are algorithm tests on fixed inputs and are kept separate from the real-video measurements reported below.

4. Calibration results per product

The same anchor procedure gives different thresholds for different products, which is the point — a mostly-white printed kurti and a dyed teal saree do not fail in the same way.

Anchor (deliberate damage)	Saree fidelity	Kurti fidelity
source (identity)	100.00	100.00
hue shift 5 deg	93.14	99.57
hue shift 10 deg	76.13	93.68
hue shift 20 deg	59.77	79.65
blur k=5	78.34	85.91
blur k=9	73.06	79.14
desaturate 0.6	83.33	90.27
Calibrated pass mark	51.0	63.2

A 10-degree hue shift costs the saree 23.9 points but the kurti only 6.3, because the kurti is largely white and neutral — a hue rotation barely moves its palette. The kurti's pass mark is therefore higher (63.2 against 51.0).

5. Results — saree run (4 videos)

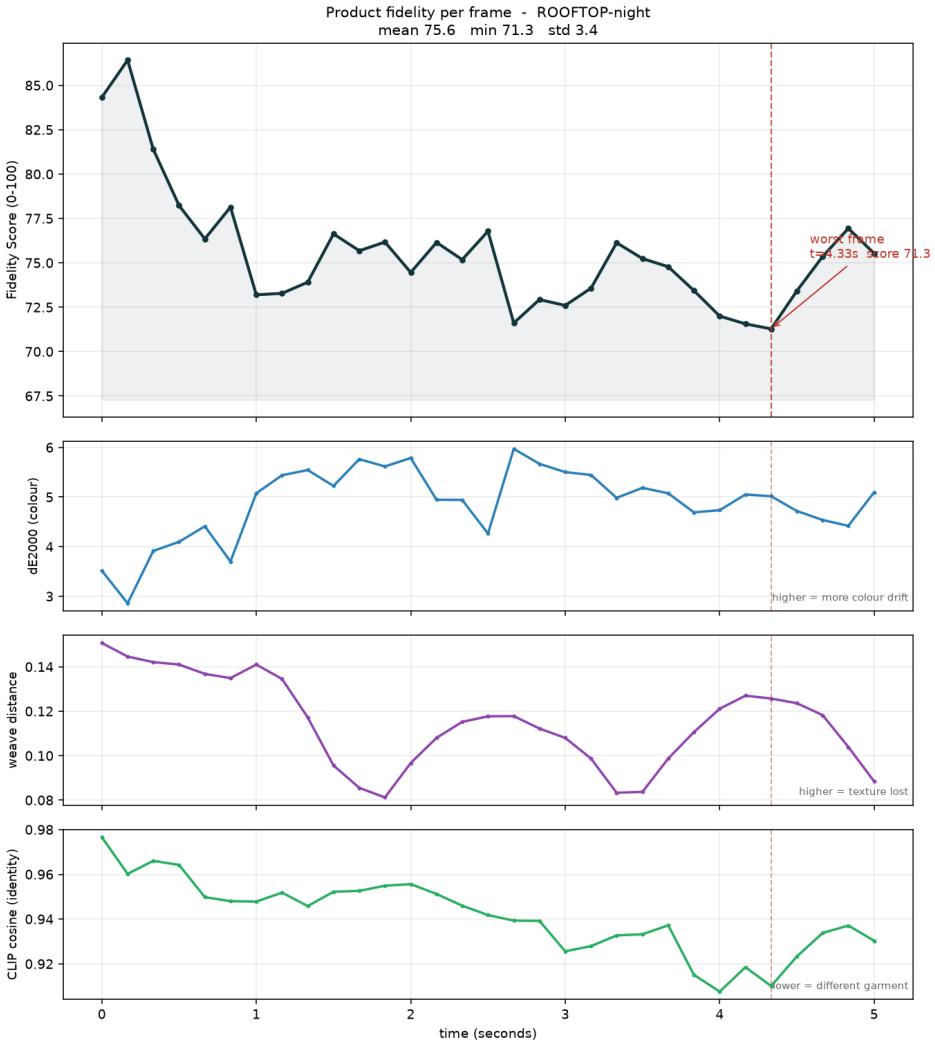
Video	Mean	Min	ΔE2000	Weave	CLIP	SSIM	Motion max	Garment %	Gate 1
ROOFTOP-night	75.57	71.28	4.88	0.115	0.941	0.219	0.052	26.1-38.4	PASS
MANDAP-dusk	71.77	65.09	5.39	0.197	0.938	0.404	0.123	12.2-38.3	PASS
BASELINE-courtyard	71.23	61.06	3.24	0.205	0.910	0.222	0.056	25.4-37.8	PASS
MANSION-marble	44.89	37.87	8.98	0.279	0.876	0.141	0.268	3.2-38.5	REJECT

Pass mark 51.0, calibrated on this product. Three videos passed; one was rejected.

6. PASS example — ROOFTOP-night

The product stays faithful for the whole clip. Its weakest single frame (71.28) still clears the pass mark of 51.0 by a wide margin, and the standard deviation of 3.39 shows the score barely moves.

Measure	Value	Reading
Fidelity mean	75.57	well above the 51.0 pass mark
Fidelity minimum	71.28	worst frame still clears the mark
Standard deviation	3.39	most stable of the four videos
Colour ΔE2000	4.88	dye held
Weave distance	0.1150	lowest texture loss of the four
CLIP identity	0.9411	same garment throughout
Motion max	0.0518	smooth, no cut or flicker
Garment occupancy	26.1% to 38.4% of frame	product stays large in frame



ROOFTOP-night: per-frame fidelity with the component metrics beneath. The curve stays flat and high for the full 5.1 seconds.

7. REJECT example — MANSION-marble

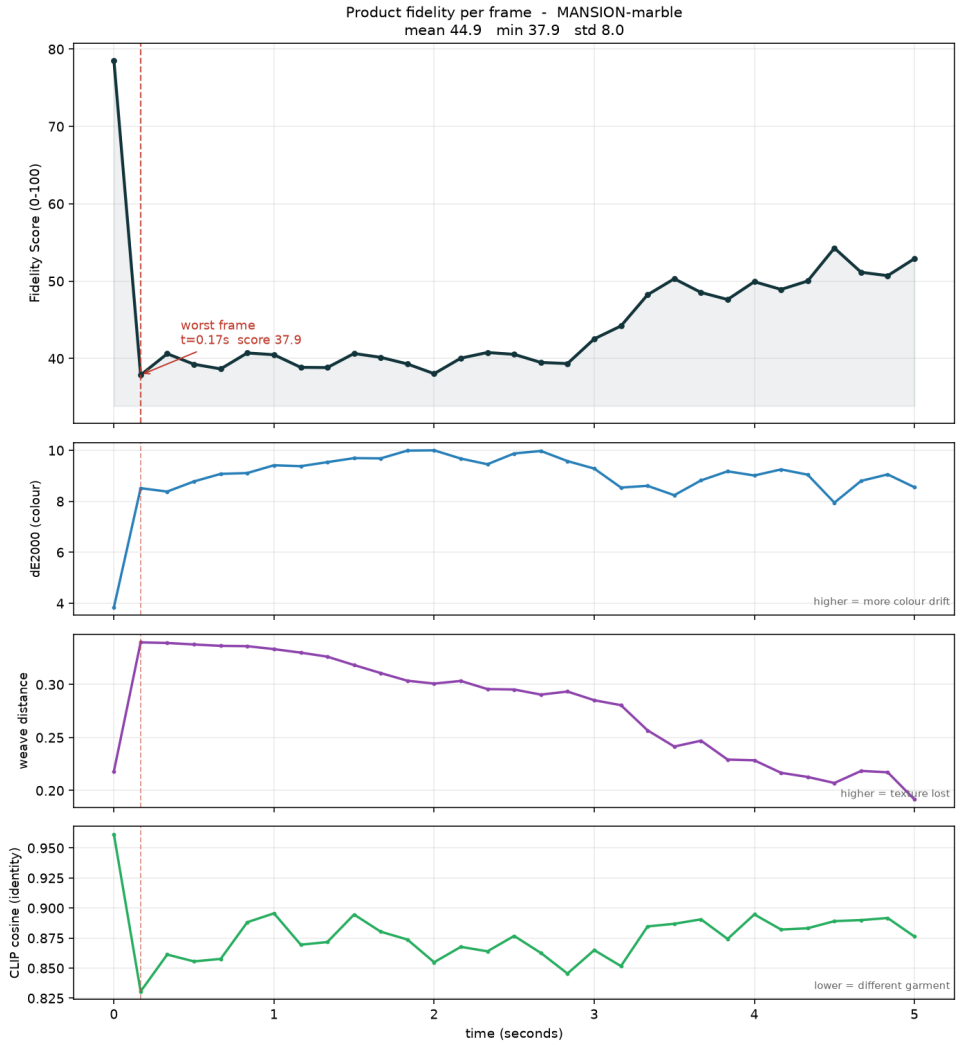
This is the main quality-testing result. The video was rejected by Gate 1 on **two independent checks**, and the rejection was confirmed by inspecting frames rather than trusted from the score alone.

Measure	Value	Verdict
Fidelity mean	44.89	below the 51.0 pass mark, and below it for essentially the whole clip
Fidelity minimum	37.87	worst frame at t = 0.17s
Motion max	0.2683	exceeds the 0.25 limit - a hard scene cut
Garment occupancy	38.5% falling to 3.2%	product collapses to a few percent of frame
Colour ΔE2000	8.98	high, but see the note below

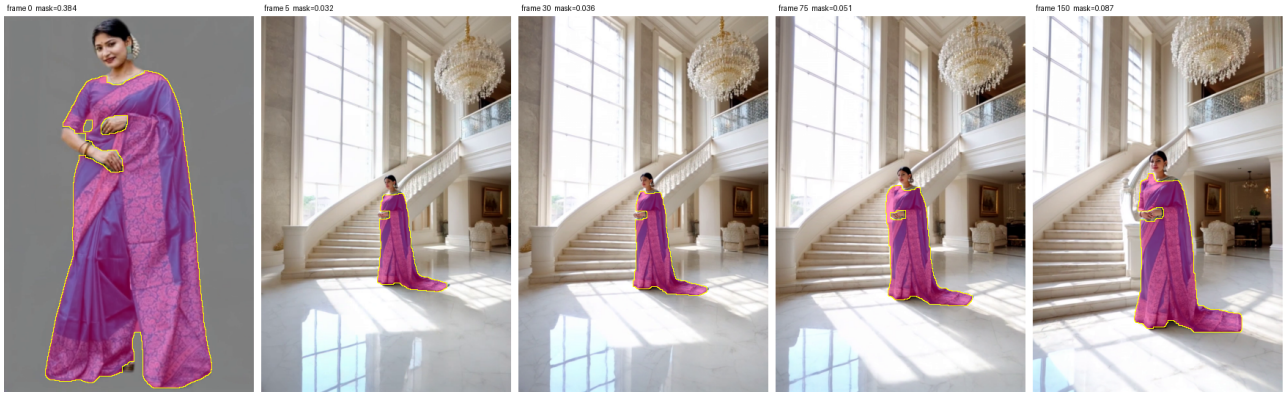
This was not primarily a dye-colour failure. The colour figure of 8.98 is elevated, but with the garment at only a few percent of the frame that number is not reliable evidence of a dye change — it largely reflects the product being tiny and washed out against bright marble. Frame inspection confirms the saree is still teal. The actual failure is **product visibility collapsing after a hard scene cut**.

Recorded Gate 1 reason, verbatim:

```
fidelity_mean 44.9 below threshold 51.0; motion spike 0.268 -- likely flicker or a cut
```



MANSION-marble: fidelity sits below the pass mark for the whole clip, unlike the transient dip seen in the passing videos.



Frame-level diagnosis. Frame 0 is a near-copy of the reference (38% of frame); after a hard cut the model is tiny in a marble hall for the remaining five seconds. The mask (pink) tracks correctly — the product really is that small.



Weave Guard for the worst frame: the motif could not be confidently located anywhere on the garment, which is itself the finding.

8. Second product type — kurti

A real kurti image (2500×2500) was analysed and a real Ekly-generated kurti video was scored through the identical pipeline. Segmentation, palette extraction and the detail-watch region all worked on a product with a completely different construction — a mostly white garment with dense printed panels rather than a dyed weave.

Measure	Kurti value	Note
Source image	upload.png, 2500x2500	square, not portrait
Garment mask	10.95% of frame	vs 35.73% for the saree
Delivered video	960x960	square 1:1, not the 9:16 requested
Fidelity mean / min	55.04 / 38.67	worst frame at t = 1.50s
Colour ΔE2000	5.00	moderate
Weave distance	0.2611	print detail partly lost
CLIP identity	0.8479	garment recognisable
Motion max	0.0407	smooth, no cut
Garment occupancy	6.6% to 8.7%	small but stable
Export	1080x1920, crop mode	100% of garment retained; conforms to 9:16

8.1 A threshold problem the testing exposed

When the operator ran this kurti through the application it was scored against a pass mark of **51.0** — the value calibrated on the **saree**, carried over in the interface. Against that number the clip passed at 55.04. Calibrating the same procedure on the kurti's own source image gives a pass mark of **63.2**, and against its own product the clip **does not pass**.

Pass mark applied	Origin	Kurti clip (55.04)	Outcome
51.0	calibrated on the saree	55.04	PASS
63.2	calibrated on the kurti	55.04	REJECT

This is reported as a finding, not corrected after the fact: the run's stored result is left exactly as it was measured. The lesson is that **the pass mark is a property of the product, not of the system**, and must be calibrated per product before its videos are judged.



Kurti garment mask. The printed panels are retained after the texture-protection fix; the plain white trousers and the handbag remain part of the subject.

9. The two-gate quality system

Gate 1 — is the product still faithful? Per-frame drift inside the garment mask, plus motion checks. Rejects a video for misrepresenting the product.
Gate 2 — are the accepted videos actually different from each other? Each video is reduced to one CLIP embedding averaged over sampled frames, and pairwise cosine distance is measured. Rejects a video for being a near-duplicate. A video must pass **both**.

	BASELINE-co	MANDAP-dusk	MANSION-mar	ROOFTOP-nig
BASELINE-courtya	-	0.1147	0.1999	0.0931
MANDAP-dusk	0.1147	-	0.1412	0.1213
MANSION-marble	0.1999	0.1412	-	0.1961
ROOFTOP-night	0.0931	0.1213	0.1961	-

Measure	Value
Diversity, all four videos	14.44
Diversity, Gate-1 passing set	10.97
Closest pair	BASELINE-courtyard / ROOFTOP-night (0.0931)
Most distinct pair	BASELINE-courtyard / MANSION-marble (0.1999)

The most important result of the whole exercise. The most visually distinct video in the set is **MANSION-marble** — the one Gate 1 rejected. Optimising for variety alone would have selected the video that misrepresents the product. This came out of the measurements, not the design, and it is why the two gates are kept independent.

Gate 2 measured real distances but discarded nothing here: three videos passed Gate 1 against a request for six, so the selection step had nothing to remove. The filtering itself has not been exercised on real video.

10. Safe Cut protection

Safe Cut trims a video to its longest continuous passing stretch, but only if at least 3.0 seconds survive. It only trims — it never regenerates, recolours or repairs — and the trimmed file is re-scored from the output rather than inheriting the parent's score.

Video	Outcome	Reason recorded
BASELINE-courtyard	not cut	every scored frame already clears the threshold; nothing to cut
MANDAP-dusk	not cut	every scored frame already clears the threshold; nothing to cut
MANSION-marble	not cut	longest passing window is 0.00s, under the 3.0s minimum for a usable ad
ROOFTOP-night	not cut	every scored frame already clears the threshold; nothing to cut

Nothing was cut, and that is the correct outcome. Three videos had nothing to remove. MANSION-marble fails throughout rather than transiently, so its longest clean window is 0.00s — the guard that stops Safe Cut being used to launder a failing video into a passing one.

11. What has not been tested

- **Only two product types** have been run on real generated video: saree and kurti. Lehenga, dress, shirt and the rest of the product vocabulary are supported in code but untested.
- **The kurti was tested with one video only.** Gate 2 could not run on it, so diversity has never been exercised on a second product.
- **Gate 2 has never rejected anything on real video.** The passing pool has always been smaller than the number of ads requested.
- **Safe Cut has never fired on real footage.** Its logic is covered by five unit tests, but no real clip has yet met the conditions for a trim.
- **No human ground truth.** The Fidelity Score has not been validated against human judgement of whether a video shows the same garment. Thresholds rest on the degradation anchors and small real samples (4 saree clips, 1 kurti clip).
- **Segmentation limits stand.** The garment mask is a heuristic refinement of a subject mask, not a trained garment segmenter, and it keeps accessories such as the kurti model's handbag.

12. Conclusion

The system does perform product consistency testing rather than asserting it. Eleven distinct checks run on every video; the metrics were validated against damage of known severity before being trusted; thresholds are derived from the product's own image rather than chosen; and the testing produced a real rejection (MANSION-marble, on two independent grounds) alongside real passes.

The testing also did what good testing is supposed to do — it found a problem in our own process. Applying the saree's pass mark to a kurti flipped that clip's verdict from PASS to REJECT once the kurti was calibrated on its own product. That is now documented in section 8.1 and is the single most important thing to fix before this is used on a wider catalogue.

Generated by `tools/build_m3_3_pdf.py` from `evidence/m33_test_results.json` and the run directories `runs/run_m2_real/` and `runs/run_20260822_183825/`.